

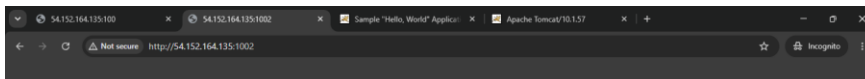
EOD Report – HAProxy, Nginx, Apache2 and Apache Tomcat

Date: 15 August 2026

Task: Web Server, Application Server and HAProxy Configuration

1. Nginx Installation and Configuration

- Installed Nginx on the server.
- Configured Nginx to run on **port 100**.
- Verified that the Nginx service is running successfully.
- Deployed a sample index.html page on Nginx.
- Verified the application through the configured port.

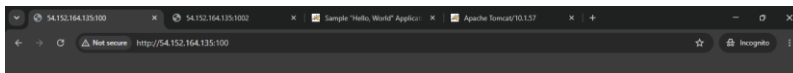


WELCOME TO MY PAGE FROM NGINX



2. Apache2 Installation and Configuration

- Installed Apache2 web server.
- Configured Apache2 to run on **port 1002**.
- Verified that the Apache2 service is active and running.
- Deployed a sample index.html page.
- Tested the Apache2 application using the configured port.



"WELCOME TO MY PAGE FROM APACHE"

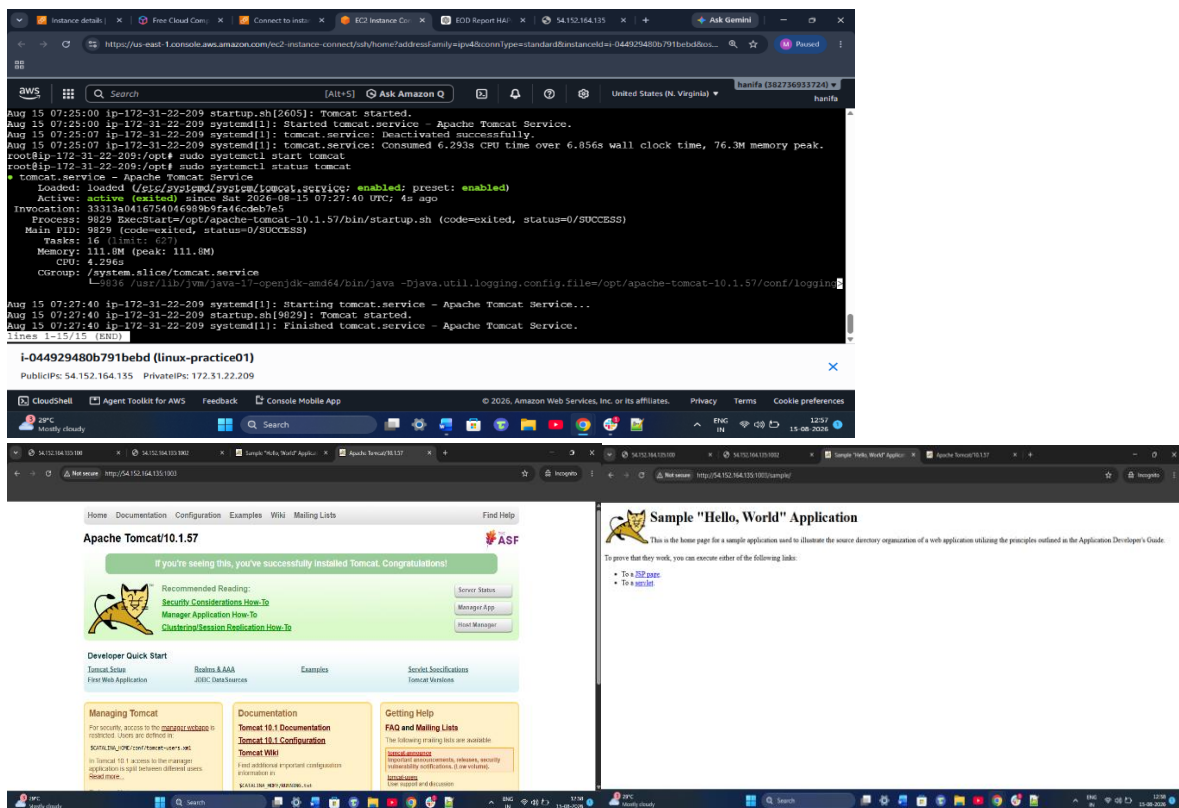


3. Apache Tomcat Installation and Configuration

- Installed Apache Tomcat.
- Configured Tomcat to run on **port 1003**.
- Verified Tomcat connectivity and service status.
- Deployed a sample application inside the Tomcat webapps directory.
- Tested the deployed application successfully.

Tomcat Systemd Service

- Created a custom tomcat.service file.
- Configured the service to manage Tomcat using systemctl.
- Tested the following operations:
 - Start Tomcat
 - Stop Tomcat
 - Restart Tomcat
 - Check Tomcat status
- Enabled Tomcat to start automatically during system boot.



4. HAProxy Configuration

- Installed and configured HAProxy.

- Configured HAProxy to act as a frontend/load-balancing layer.
- Configured backend servers for the Nginx, Apache2 and Tomcat services.
- Verified traffic forwarding from HAProxy to the configured backend services.
- Tested the setup to ensure the web servers and application server were accessible through HAProxy.

The screenshot shows an AWS CloudShell terminal window with the following content:

```

• haproxy.service - HAProxy Load Balancer
  Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
  Active: active (running) since Sat 2026-08-15 07:28:42 UTC; 5min ago
  Invocation: c041fef89a614b96a823041c9c38f53b
  Docs: man:haproxy(1)
        file:/usr/share/doc/haproxy/configuration.txt.gz
  Main PID: 16365 (haproxy)
  Status: "Ready."
  Tasks: 3 (limit: 627)
  Memory: 45.3M (peak: 47.6M)
  CPU: 222ms
  CGroup: /system.slice/haproxy.service
          └─16365 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
          └─16426 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock

Aug 15 07:28:42 ip-172-31-22-209 systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
Aug 15 07:28:42 ip-172-31-22-209 haproxy[16365]: [NOTICE] (16365) : Initializing new worker (16426)
Aug 15 07:28:42 ip-172-31-22-209 haproxy[16365]: [NOTICE] (16365) : Loading success.
Aug 15 07:28:42 ip-172-31-22-209 systemd[1]: Started haproxy.service - HAProxy Load Balancer.
Aug 15 07:28:44 ip-172-31-22-209 haproxy[16426]: [WARNING] (16426) : Server webserver/tomcat is DOWN, reason: Layer4 connection problem
Aug 15 07:28:44 ip-172-31-22-209 haproxy[16426]: Server webserver/tomcat is DOWN, reason: Layer4 connection problem, info: "Connection refused"
lines 1-21
  
```

Below the terminal output, the instance details are shown: **i-044929480b791bebd (linux-practice01)**, with Public IPs: 54.152.164.135 and Private IPs: 172.31.22.209.

Conclusion

Successfully completed the setup and configuration of **Nginx, Apache2, Apache Tomcat and HAProxy**. Nginx was configured on port **80**, Apache2 on port **100**, and Tomcat on port **1002**. Sample web/application content was deployed and a custom systemd service was created for Tomcat. HAProxy was configured to route traffic to the backend services and the complete setup was tested successfully.